Here the average of $\sin^2 \omega t$ over a cycle is $1/2$ and that of $\sin \omega t \cos \omega t$ is zero. Substituting these values, we obtain average power over a cycle.

$$P_{av} = V_m I_m \cos\phi \times 1/2$$

$$= \frac{V_m}{\sqrt{2}} \frac{I_m}{\sqrt{2}} \cos\phi$$

$$P_{av} = V_{RMS}\ I_{RMS} \cos\phi \qquad (4.62)$$

where $V_{RMS} I_{RMS}$ is called apparent power and cos ϕ is power factor. The average power of an AC circuit is also known as the true power of the circuit.

Special Cases

(i) For a purely resistive circuit, the phase angle between voltage and current is zero and cos $\phi = 1$.
$\therefore P_{av} = V_{RMS}\ I_{RMS}$

(ii) For a purely inductive or capacitive circuit, the phase angle is $\pm \pi/2$ and $\cos\left(\pm \pi/2\right) = 0$.
$\therefore P_{av} = 0$

(iii) For series RLC circuit, the phase angle $\phi = \tan^{-1}\left(\frac{X_L - X_C}{R}\right)$
$\therefore P_{av} = V_{RMS}\ I_{RMS} \cos\phi$

(iv) For series RLC circuit at resonance, the phase angle is zero and $\cos\phi = 1$.
$\therefore P_{av} = V_{RMS}\ I_{RMS}$

4.8.2 Wattless current

Consider an AC circuit in which there is a phase angle of ϕ between V_{RMS} and I_{RMS} and voltage is assumed to be leading the current by ϕ as shown in the phasor diagram (Figure 4.55).

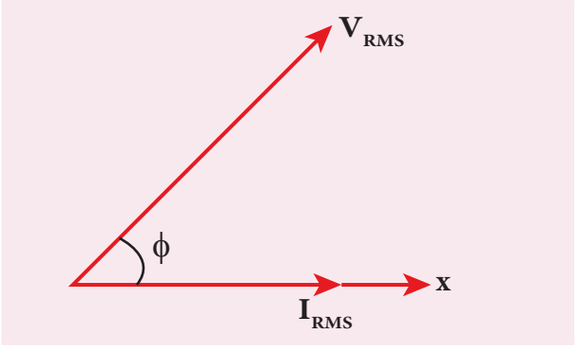


Figure 4.55 V_{RMS} leads I_{RMS} by ϕ

Now, I_{RMS} is resolved into two perpendicular components namely, $I_{RMS} \cos\phi$ along V_{RMS} and $I_{RMS} \sin\phi$ perpendicular to V_{RMS} as shown in Figure 4.56.

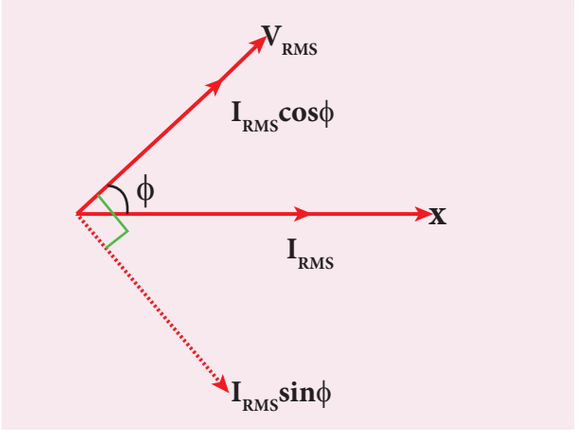


Figure 4.56 The components of I_{RMS}

(i) The component of current $\left(I_{RMS} \cos\phi\right)$ which is in phase with the voltage is called active component. The power consumed by this current $= V_{RMS} I_{RMS} \cos\phi$. So that it is also known as 'Wattful' current.

(ii) The other component $\left(I_{RMS} \sin\phi\right)$ which has a phase angle of $\pi/2$ with the voltage is called reactive component. The power consumed is zero. So that it is also known as 'Wattless' current.

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